

KALUA CLI Reference

USER_GUIDE.md §3 | A4 Landscape

Commands

check app.lua

Static validation

Syntax, unknown k.*, main()

+ gofmt-style formatter

run app.lua

Interactive web app

127.0.0.1:9000

Per-tab Lua session

new name

Scaffold app.lua

lsp

Language Server

(stdio, UTF-8)

serve app.lua

Headless API

HTTP/WS/TCP

Worker pool + k.shared.*

mcp

MCP Server

(stdio, 9 tools)

builder app.lua

Visual form editor

Multi-form .lua/.json

AI chat panel

ai gen|fix|val

NL → Lua

generate/fix/validate

version

Print version

-l, -

-p, --por

-V, --ve

--repl-o

-a, --ar

--de

--test

--watch

-d, --db M

-f, --allo

run flags

--session-limit N (8)

rt N (9000, 0=ephemeral)

erbose (trace k.* calls)

--ini PATH

on-error (REPL at crash)

-n, --no-browser

rg K=V (→ ARGS table)

debug (stub, not impl)

t (headless, no HTTP)

i (poll 400ms, hot reload)

NAME=DSN (repeatable)

ow-fs PATH (repeatable)

serve flags

-w, --workers N (4)

--host H (127.0.0.1)

--debug/--debug-worker (stub)

-d/--db, -a/--arg, -f/--allow-fs

-p, --port N (8080)

--ini PATH

-m, --mode http|ws|tcp (csv)

-v, --verbose

Signals:
SIGTERM/INT → shutdown()
SIGHUP → hot reload
(atomic worker swap)

check / formatter

-w → write in place (perm)

Validation (default):
syntax, unknown k.*, missing main

Priority: -w > -d > -l > --format
Single file: flag anywhere
Multi-file: flag first

Rules: 2-space indent, normalize spaces
preserve comments/strings, idempotent
re-parses output

--format → stdout (no write)

[CHECK] in KALUA.INI:
format, w, l, d, verbose

-l files... → list unformatted

-d → unified diff

n

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builder • ai • INI

--model NAME

-n, --no-browser

generate \"req\" [-o file]

Format: [SECTION] key=value
Sections: RUN, SERVE, BUILDER, CHECK, AI
Keys=long flags (case-ins, -/_=)
Bool: 1/true/yes/on | 0/false/no/off
Bare key=true, repeat → append

fix app.lua

Example: KALUA.ini.sample
[RUN] no-browser=1 port=9000 watch=1
[AI] KALUA_AI_BASE_URL=...
KALUA_AI_API_KEY=...
KALUA_AI_MODEL=...

-p, --port N (9001)

INI Search (1st wins):
1. --ini path
2. KALUA_INI env
3. ./KALUA.INI
Missing default=OK, explicit missing=exit 3

validate app.lua

--ini PATH

Features: multi-form v3, source-preserving, AST import (line spans), verbatim handlers, v2 → v3 migration, AI chat (/api/ai/stream), Tabulator proxy, tabbed modal

Config precedence:
1. CLI flags
2. [AI] in INI
3. KALUA_AI_* env
4. OPENAI_API_KEY (OpenRouter)

Precedence: CLI > INI > env > defaults
INI only fills unset CLI flags

Exit Codes • Restrictions

2 — CLI Usage Error

0 — OK

run: UI bindings only (forms, controls)
serve: handle_http/ws/tcp + k.shared.*
UI in serve → runtime error

Must define function main()
(entry point for both modes)

1 — Script/Runtime Error

Hot reload: run --watch / SIGHUP
400ms poll + SHA256 dedupe
Broken script → keeps old app

Flags before OR after script arg
--ini works for all commands
ARGS from --arg K=V

3 — File I/O Error

Sandbox: no require, print, dofile,
loadfile, os.execute, io, os.getenv
Use k. API (k.print, k.file_)

DSN: sqlite:// mysql://
postgres:// sqlserver://
--db NAME=DSN →
k.connect_db(\"#NAME\")

